

# CM0832 - MT5001 Elements of Set Theory

## 1. Sets

Andrés Sicard-Ramírez

Universidad EAFIT

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# Outline

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Preliminaries

First-Order Logic

The ZFC Theory

Properties

The Axioms

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# Preliminaries

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## Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

## Acronyms

ZFC Zermelo-Fraenkel set theory with Choice

FOL First-Order Logic

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# First-Order Logic

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## Theorem

The identity (equality) relation is an equivalence relation.

# First-Order Logic

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## Theorem

The identity (equality) relation is an equivalence relation.

That is, for all  $X, Y, Z$ ,

- (i)  $X \doteq X$  (reflexivity),
- (ii) if  $X \doteq Y$ , then  $Y \doteq X$  (symmetry),
- (iii) if  $X \doteq Y$  and  $Y \doteq Z$ , then  $X \doteq Z$  (transitivity).

# First-Order Logic

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## Definition

A **proposition** (or **statement**) is a sentence that can be assigned a truth value of **true** or **false**.

# First-Order Logic

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## Definitions

A **propositional function** (or **property**) is a sentence containing one or more variables (parameters) that becomes a proposition when specific values are assigned to its variables.

# First-Order Logic

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## Definitions

A **propositional function** (or **property**) is a sentence containing one or more variables (parameters) that becomes a proposition when specific values are assigned to its variables.

The **domain** of a propositional function is the **domain of the discourse**.

The **codomain** of a propositional function is  $\{\text{true}, \text{false}\}$ .

# First-Order Logic

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## Theorem

The identity (equality) relation is substitutive.

# First-Order Logic

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## Theorem

The identity (equality) relation is substitutive.

That is, let  $\mathbf{P}(X)$  be a propositional function. For all  $X, Y$ , if  $X \doteq Y$  and  $\mathbf{P}(X)$ , then  $\mathbf{P}(Y)$ .

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# Pure Sets

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## Definition

A set is **pure** if and only its members are also sets.

# Pure Sets

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## Definition

A set is **pure** if and only its members are also sets.

## Description

The ZFC theory is a first-order theory of pure sets.

# First-Order Language of ZFC

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## Description

In our informal but axiomatic presentation of ZFC we have two primitive (undefined) concepts: **set** and **membership relation**.

# First-Order Language of ZFC

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## Description

In our informal but axiomatic presentation of ZFC we have two primitive (undefined) concepts: **set** and **membership relation**.

The (binary) membership relation is denoted by  $\in$ .

# First-Order Language of ZFC

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## Notation

$$X \neq Y \stackrel{\text{def}}{=} \neg(X = Y),$$

$$X \notin Y \stackrel{\text{def}}{=} \neg(X \in Y).$$

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# Properties

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## Example

We define the following unary property:

$\mathbf{P}(X)$ : “There exists no  $Y \in X$ ”.

# Properties

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## Example

We define the following unary property:

$\mathbf{P}(X)$ : “There exists no  $Y \in X$ ”.

Since that we can prove that there is a **unique** set  $X$  with the property  $\mathbf{P}(X)$  we can **add** a new **constant** symbol  $\emptyset$  denoting the empty set.

# Properties

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## Example

We define the following binary property:

$P(X, Y)$ : “Every element of  $X$  is an element of  $Y$ ”.

# Properties

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## Example

We define the following binary property:

$\mathbf{P}(X, Y)$ : “Every element of  $X$  is an element of  $Y$ ”.

Using the above property we can **add** a new binary **relation** symbol  $\subseteq$  denoting that  $X$  is a subset of  $Y$ :

$$X \subseteq Y \stackrel{\text{def}}{=} \mathbf{P}(X, Y).$$

# Properties

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## Example

We define the following ternary property:

$Q(X, Y, Z)$ : “For every  $U$ ,  $U \in Z$  if and only if  $U \in X$  and  $U \in Y$ ”.

# Properties

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## Example

We define the following ternary property:

$Q(X, Y, Z)$ : “For every  $U$ ,  $U \in Z$  if and only if  $U \in X$  and  $U \in Y$ ”.

Since that we can prove that for every  $X$  and  $Y$  there is a **unique**  $Z$  such that  $Q(X, Y, Z)$ , we can **add** a new binary **function** symbol  $\cap$ , denoting the intersection of  $X$  and  $Y$ :

$$X \cap Y \stackrel{\text{def}}{=} \text{the unique } Z \text{ such that } Q(X, Y, Z).$$

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# The Axioms

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## The Axiom of Existence

- “*There exists a set which has no elements.*” (p. 7)
- $(\exists B)(\forall x)(x \notin B).$

# The Axioms

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## The Axiom of Existence

- “*There exists a set which has no elements.*” (p. 7)
- $(\exists B)(\forall x)(x \notin B).$

### Remark

The Axiom of Existence postulate the our universe of discourse is not void. This axiom postulates the existence of the empty set.

# The Axioms

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## The Axiom of Existence

- “*There exists a set which has no elements.*” (p. 7)
- $(\exists B)(\forall x)(x \notin B).$

### Remark

The Axiom of Existence postulate the our universe of discourse is not void. This axiom postulates the existence of the empty set.<sup>†</sup>

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<sup>†</sup>Strictly speaking, the universe of the discourse of FOL is not empty. For example, let  $\mathbf{P}(x)$  be a property, then  $(\forall x)\mathbf{P}(x) \rightarrow (\exists x)\mathbf{P}(x)$  is a theorem. See, e.g. (Lambert 2001).

# The Axioms

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## The Axiom of Extensionality

- “If every element of  $X$  is an element of  $Y$  and every element of  $Y$  is an element of  $X$ , then  $X \doteq Y$ .” (p. 7)
- “If two sets have exactly the same members, then they are equal.” (Enderton 1977)
- $(\forall X)(\forall Y)[(\forall z)(z \in X \leftrightarrow z \in Y) \rightarrow X \doteq Y]$ .

# The Axioms

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## Lemma [1.]3.1

There exists only one set with no elements.

# The Axioms

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## Lemma [1.]3.1

There exists only one set with no elements.

## Proof

On the whiteboard.

# The Axioms

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## Lemma [1.]3.1

There exists only one set with no elements.

### Proof

On the whiteboard.

## Definition [1.]3.2

The **empty set**, denoted  $\emptyset$ , is set with no elements.

# The Axioms

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## Theorem (converse of the Axiom of Extensionality)

If  $X \doteq Y$ , then every element of  $X$  is an element of  $Y$  and every element of  $Y$  is an element of  $X$ .

# The Axioms

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## Theorem (converse of the Axiom of Extensionality)

If  $X \doteq Y$ , then every element of  $X$  is an element of  $Y$  and every element of  $Y$  is an element of  $X$ .

$$(\forall X)(\forall Y)[X \doteq Y \rightarrow (\forall z)(z \in X \leftrightarrow z \in Y)].$$

**Proof**

FOL.

# The Axioms

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## Theorem (converse of the Axiom of Extensionality)

If  $X \doteq Y$ , then every element of  $X$  is an element of  $Y$  and every element of  $Y$  is an element of  $X$ .<sup>†</sup>

$$(\forall X)(\forall Y)[X \doteq Y \rightarrow (\forall z)(z \in X \leftrightarrow z \in Y)].$$

## Proof

FOL.

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<sup>†</sup>Due to this theorem some authors (see, e.g. (Goldrei 1996, p. 76)) state the Axiom of Extensionality using a bi-conditional, that is, if every element of  $X$  is an element of  $Y$  and every element of  $Y$  is an element of  $X$  if and only if  $X \doteq Y$ .  $(\forall X)(\forall Y)[(\forall z)(z \in X \leftrightarrow z \in Y) \leftrightarrow X \doteq Y]$ .

# The Axioms

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## Theorem

The identity (equality) relation is compatible with the membership relation.

# The Axioms

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## Theorem

The identity (equality) relation is compatible with the membership relation.

That is, for all  $X, Y, Z$ ,

- (i) if  $X \doteq Y$  and  $X \in Z$ , then  $Y \in Z$ ,
- (ii) if  $X \doteq Y$  and  $Z \in X$ , then  $Z \in Y$ .

# The Axioms

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## Theorem

The identity (equality) relation is compatible with the membership relation.

That is, for all  $X, Y, Z$ ,

- (i) if  $X \doteq Y$  and  $X \in Z$ , then  $Y \in Z$ ,
- (ii) if  $X \doteq Y$  and  $Z \in X$ , then  $Z \in Y$ .

## Proof

- (i) Hint. To use that the identity is substitutive and the property  $\mathbf{P}(A, B)$ : “ $A \in B$ ”.
- (ii) Hint. To use the converse of the Axiom of Extensionality.

# The Axioms

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## The Axiom Schema of Comprehension

- “Let  $\mathbf{P}(x)$  be a unary property of  $x$ . For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .” (p. 8)
- $(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x))$ .

# The Axioms

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## The Axiom Schema of Comprehension

- “Let  $\mathbf{P}(x)$  be a unary property of  $x$ . For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .” (p. 8)
- $(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x))$ .

### Remark

We postulate an axiom **schema**, i.e., there is an axiom for **each** property  $\mathbf{P}(x)$ .

# The Axioms

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## The Axiom Schema of Comprehension

- “Let  $\mathbf{P}(x)$  be a unary property of  $x$ . For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .” (p. 8)
- $(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x)).$

## Example

Let  $\mathbf{P}(x)$ : “ $x \notin x$ ”. The axiom postulates:

For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $x \notin x$  (the set  $B$  is the empty set).

# The Axioms

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## The Axiom Schema of Comprehension

- “Let  $\mathbf{P}(x)$  be a unary property of  $x$ . For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .” (p. 8)
- $(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x))$ .

## Remark

*“Although the supply of axioms is unlimited, this causes no problems, since it is easy to recognize whether a particular statement is or is not an axiom and since every proof uses only finitely many axioms.” (p. 8)*

# The Axioms

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## The Axiom Schema of Comprehension

- “Let  $\mathbf{P}(x)$  be a unary property of  $x$ . For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .” (p. 8)
- $(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x))$ .

### Remark

When using the axiom scheme with a  $(n + 1)$ -ary property  $\mathbf{P}(x, p_1, \dots, p_n)$  the axiom is:

For any sets  $p_1, \dots, p_n, A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x, p_1, \dots, p_n)$ .

$$(\forall p_1) \cdots (\forall p_n)(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x, p_1, \dots, p_n)).$$

# The Axioms

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## Example [1.]3.3

To prove that if  $A$  and  $B$  are sets, then there is a set  $C$  such that  $x \in C$  if and only if  $x \in A$  and  $x \in B$ .

# The Axioms

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## Example [1.]3.3

To prove that if  $A$  and  $B$  are sets, then there is a set  $C$  such that  $x \in C$  if and only if  $x \in A$  and  $x \in B$ .

## Proof

Hint: To define the binary property  $\mathbf{P}(x, B)$ : “ $x \in B$ ” and use the Axiom Schema of Comprehension.

# The Axioms

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## Question

Let  $\mathbf{P}(x)$ : “ $x \notin x$ ”. Without using the Axiom of Existence can we prove the existence of the empty set from the Axiom Schema of Comprehension and the property  $\mathbf{P}(x)$ ?

# The Axioms

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## Lemma [1.]3.4

For every  $A$ , there is only one set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .

# The Axioms

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## Lemma [1.]3.4

For every  $A$ , there is only one set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .

## Remark

This lemma establishes that the set postulated by the Axiom Schema of Comprehension is **unique**.

# The Axioms

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## Lemma [1.]3.4

For every  $A$ , there is only one set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .

### Proof

On the whiteboard.

# The Axioms

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## Lemma [1.]3.4

For every  $A$ , there is only one set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .

## Question

Given the above lemma, why not state the Axiom Schema of Comprehension by “for any set  $A$ , there is a **unique** set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ ”?

$$(\forall A)(\exists! B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x)).$$

# The Axioms

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## Definition [1.]3.5

Given a set  $A$  and a property  $\mathbf{P}(x)$ , the set  $B$  postulated by the Axiom Schema of Comprehension is

$$\{ x \in A \mid \mathbf{P}(x) \}.$$

# The Axioms

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## Definition [1.]3.5

Given a set  $A$  and a property  $\mathbf{P}(x)$ , the set  $B$  postulated by the Axiom Schema of Comprehension is

$$\{ x \in A \mid \mathbf{P}(x) \}.\dagger$$

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<sup>†</sup>Strictly speaking before this definition we should introduce the abstraction symbol  $\{ \cdot \mid \cdot \}$  in the language of ZFC. See, e.g. (Drake 1974).

# The Axioms

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## The Axiom of Pair

- “For any  $A$  and  $B$ , there is a set  $C$  such that  $x \in C$  if and only if  $x \doteq A$  or  $x \doteq B$ .” (p. 9)
- “For any sets  $A$  and  $B$ , there is a set having as members just  $A$  and  $B$ .” (Enderton 1977)
- $(\forall A)(\forall B)(\exists C)(\forall x)(x \in C \leftrightarrow x \doteq A \vee x \doteq B).$

# The Axioms

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## The Axiom of Pair

- “For any  $A$  and  $B$ , there is a set  $C$  such that  $x \in C$  if and only if  $x \doteq A$  or  $x \doteq B$ .” (p. 9)
- “For any sets  $A$  and  $B$ , there is a set having as members just  $A$  and  $B$ .” (Enderton 1977)
- $(\forall A)(\forall B)(\exists C)(\forall x)(x \in C \leftrightarrow x \doteq A \vee x \doteq B).$

## Exercise

To prove that the set postulated by the axiom is unique.

# The Axioms

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## Definition

The **unordered pair** of  $A$  and  $B$ , denoted by  $\{A, B\}$ , is the set postulated by the Axiom of Pair.

# The Axioms

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## Definition

The **unordered pair** of  $A$  and  $B$ , denoted by  $\{A, B\}$ , is the set postulated by the Axiom of Pair.

## Notation

If  $A \doteq B$  then  $\{A\} \stackrel{\text{def}}{=} \{A, A\}$ .

# The Axioms

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## The Axiom of Union

- “For any set  $S$ , there exists a set  $U$  such that  $x \in U$  if and only if  $x \in A$  for some  $A \in S$ .” (p. 9)
- “For any set  $S$ , there exists a set  $U$  whose elements are exactly the members of the members of  $A$ ”. (Enderton 1977)
- $(\forall S)(\exists U)(\forall x)[x \in U \leftrightarrow (\exists A)(x \in A \wedge A \in S)]$ .

# The Axioms

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## The Axiom of Union

- “For any set  $S$ , there exists a set  $U$  such that  $x \in U$  if and only if  $x \in A$  for some  $A \in S$ .” (p. 9)
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## Exercise

To prove that the set postulated by the axiom is unique.

# The Axioms

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## The Axiom of Union

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- “For any set  $S$ , there exists a set  $U$  whose elements are exactly the members of the members of  $A$ ”. (Enderton 1977)
- $(\forall S)(\exists U)(\forall x)[x \in U \leftrightarrow (\exists A)(x \in A \wedge A \in S)]$ .

## Definition

The (**generalised**) **union** of  $S$ , denoted by  $\bigcup S$ , is the set postulated by the axiom.

# The Axioms

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## Examples

- Let  $A, B$  sets. Then,  $x \in \bigcup\{A, B\}$  if and only if  $x \in A$  or  $x \in B$ .

# The Axioms

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## Examples

- Let  $A, B$  sets. Then,  $x \in \bigcup\{A, B\}$  if and only if  $x \in A$  or  $x \in B$ .
- $\bigcup \emptyset \doteq \emptyset$ .

# The Axioms

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## Examples

- Let  $A, B$  sets. Then,  $x \in \bigcup\{A, B\}$  if and only if  $x \in A$  or  $x \in B$ .
- $\bigcup \emptyset \doteq \emptyset$ .

## Definition

The **union** of  $A$  and  $B$ , denoted by  $A \cup B$  is the set  $\bigcup\{A, B\}$ .

# The Axioms

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## Definition [1.]3.10

$A$  is a **subset** of  $B$ , denoted by  $A \subseteq B$  if and only every element of  $A$  belongs to  $B$ .

# The Axioms

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## Definition [1.]3.10

$A$  is a **subset** of  $B$ , denoted by  $A \subseteq B$  if and only every element of  $A$  belongs to  $B$ .

## Example [1.]3.11

- (a)  $\emptyset \subseteq A$  and  $A \subseteq A$  for every set  $A$ .
- (b)  $\{x \in A \mid \mathbf{P}(x)\} \subseteq A$ .
- (c) If  $A \in S$ , then  $A \subseteq \bigcup S$ .

# The Axioms

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## The Axiom of Power Set

- “For any set  $S$ , there exists a set  $P$  such that  $X \in P$  if and only if  $X \subseteq S$ .” (p. 10)
- “For any set  $S$ , there is a set whose members are exactly the subsets of  $S$ .”  
(Enderton 1977)
- $(\forall S)(\exists P)(\forall X)[X \in P \leftrightarrow X \subseteq S]$ .

# The Axioms

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## The Axiom of Power Set

- “For any set  $S$ , there exists a set  $P$  such that  $X \in P$  if and only if  $X \subseteq S$ .” (p. 10)
- “For any set  $S$ , there is a set whose members are exactly the subsets of  $S$ .”  
(Enderton 1977)
- $(\forall S)(\exists P)(\forall X)[X \in P \leftrightarrow X \subseteq S]$ .

## Exercise

To prove that the set postulated by the axiom is unique.

# The Axioms

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## The Axiom of Power Set

- “For any set  $S$ , there exists a set  $P$  such that  $X \in P$  if and only if  $X \subseteq S$ .” (p. 10)
- “For any set  $S$ , there is a set whose members are exactly the subsets of  $S$ .”  
(Enderton 1977)
- $(\forall S)(\exists P)(\forall X)[X \in P \leftrightarrow X \subseteq S]$ .

## Definition

The **power set** of  $S$ , denoted by  $\mathcal{P}(S)$ , is the set postulated by the axiom, i.e., is the set of all subsets of  $S$ .

# The Axioms

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## Convention (p. 11)

If it has been proved that some set  $A$  contains all  $x$  with the property  $\mathbf{P}(x)$  then

$$\{ x \mid \mathbf{P}(x) \} \stackrel{\text{def}}{=} \{ x \in A \mid \mathbf{P}(x) \}.$$

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# Elementary Operations on Sets

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## Definitions

$$A \subset B \stackrel{\text{def}}{=} A \subseteq B \text{ and } A \neq B$$

**(proper subset)**

$$A \supseteq B \stackrel{\text{def}}{=} A \subseteq B$$

**(superset)**

$$A \supset B \stackrel{\text{def}}{=} A \subset B$$

**(proper superset)**

# Elementary Operations on Sets

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## Definitions

Let  $A$  and  $B$  sets. Then

$$A \cup B \stackrel{\text{def}}{=} \bigcup \{A, B\}$$

**(union)**

$$A \cap B \stackrel{\text{def}}{=} \{x \in A \mid x \in B\}$$

**(intersection)**

$$A - B \stackrel{\text{def}}{=} \{x \in A \mid x \notin B\}$$

**(difference)**

$$A \triangle B \stackrel{\text{def}}{=} (A - B) \cup (B - A)$$

**(symmetric difference)**

# Elementary Operations on Sets

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## Convention (pp. 9–10)

*“We say that  $S$  is a **system of sets** or a **collection of sets** when we want to stress that elements of  $S$  are sets (of course, this is always true—all our objects are sets—and thus the expressions ‘set’ and ‘system of sets’ have the same meaning).”*

# Elementary Operations on Sets

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## Exercise

To prove that for any **non-empty** system of sets  $S$ , there exists a unique set  $B$  such that  $x \in B$  if and only if  $x$  belongs to every member of  $S$ .

# Elementary Operations on Sets

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## Exercise

To prove that for any **non-empty** system of sets  $S$ , there exists a unique set  $B$  such that  $x \in B$  if and only if  $x$  belongs to every member of  $S$ .

## Proof

- (i) We prove the existence of the set  $B$  from the hypothesis that exists  $A \in S$  and the Axiom Schema of Comprehension on property  $\mathbf{P}(x, S)$ : “ $x$  belongs to every member of  $S$ ”.
- (ii) We prove that the set  $B$  is unique from the Axiom of Extensionality.

# Elementary Operations on Sets

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## Exercise

To prove that for any **non-empty** system of sets  $S$ , there exists a unique set  $B$  such that  $x \in B$  if and only if  $x$  belongs to every member of  $S$ .

## Definition

The **(generalised) intersection** of a **non-empty** system of sets  $S$ , denoted by  $\bigcap S$ , is the set postulated by the exercise.

# Elementary Operations on Sets

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## Question

Note that  $\bigcap \emptyset$  is not defined. Why?

# Elementary Operations on Sets

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## Question

Note that  $\bigcap \emptyset$  is not defined. Why?

## Alternative definition

$$A \cap B \stackrel{\text{def}}{=} \bigcap \{A, B\}.$$

# Elementary Operations on Sets

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## Definitions

Sets  $A$  and  $B$  are **disjoint** if and only if  $A \cap B \doteq \emptyset$ .

# Elementary Operations on Sets

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## Definitions

Sets  $A$  and  $B$  are **disjoint** if and only if  $A \cap B \doteq \emptyset$ .

A system of sets  $S$  is **mutually disjoint** (or **pairwise disjoint**) if and only if  $A \cap B \doteq \emptyset$ , for all  $A, B \in S$  with  $A \neq B$ .

# Outline

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Preliminaries

First-Order Logic

The ZFC Theory

Properties

The Axioms

Elementary Operations on Sets

References

# References

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